Four datasets. Different winners.

Tree-model case study | One-page synthesis

CatBoost stands out on categorical IDs; XGBoost and LightGBM remain strong across mixed and numerical data. No model wins every quality and speed comparison.

Dataset and task	Data and workflow	Observed test finding
Home Credit Payment difficulty	307,511 original rows; mixed numerical/categorical features, missingness and engineered ratios. Full/filtered features; one-hot/native; baseline + tuned, ~30 min per study.	**Top boosting models closely matched.** One-hot AUC: XGBoost 0.7636; CatBoost and LightGBM 0.7635. CatBoost leads AP (0.2554). Native handling shows no clear AUC advantage.
Adult Income Income above $50K	32,537 rows after deduplication; 5 numeric + 7 categorical predictors. Redundant education text and fnlwgt removed. Tuned models, ~10 min each.	**LightGBM has the highest test AUC: 0.9231.** XGBoost is close at 0.9228; their paired AUC interval includes zero. CatBoost: 0.9211. Categories alone do not ensure a CatBoost win.
Amazon Access Access approval	32,769 rows; 9 categorical ID predictors with many distinct values. RF one-hot; boosting native. Tuned models, ~10 min each.	**CatBoost has the clearest quality advantage:** AUC 0.8809 versus LightGBM 0.8607 and XGBoost 0.8536. Log loss is 6.6% lower than LightGBM; paired loss intervals support the advantage.
HIGGS shards Signal/background	Large numerical experiment; 28 predictors, 500,000 validation + 500,000 test rows. TFRecord to NumPy. Three 100-tree baselines; no tuning; RF excluded.	**XGBoost leads quality:** AUC 0.8135 versus LightGBM 0.8100 and CatBoost 0.8008. LightGBM trains fastest, but only 3.8% faster than XGBoost.

Training speed and inference are different outcomes

Experiment	Fastest recorded fit	Fastest observed inference
Home Credit	LightGBM (one-hot finalists)	CatBoost one-hot; thread settings need audit
Adult	XGBoost: 1.19 s	CatBoost: 8.8 ms / 1,000; XGBoost close at 9.5 ms
Amazon	LightGBM: 1.68 s	CatBoost: 19.90 ms / 1,000 including preparation
HIGGS	LightGBM: 92.59 s; XGBoost: 96.25 s	CatBoost: 0.432 ms / 1,000; 2.14x XGBoost throughput

What changed across datasets?

Categorical structure: Amazon exposes a strong CatBoost advantage; Adult does not. **Mixed credit data:** engineering, missingness and representation choices leave the leading boosting models close. **Large numerical data:** HIGGS favors XGBoost for quality and LightGBM modestly for fitting speed. Dataset size alone does not explain these differences.

Scope: These are within-dataset results, not a controlled test of dataset size or categorical handling alone. Budgets, tuning objectives and model settings differ; HIGGS is untuned. AUC values across different tasks do not rank task difficulty. Inference protocols differ (Amazon includes preparation); compare timings within each study. Single splits and mostly single fit timings limit generalization.

Sources: Home Credit Findings; Adult Income Findings Explained; Amazon Employee Access Results / Inference Report; HIGGS Baseline Results. Only the four completed experiments are summarized.